

Heitor Baldo, PhD

Applied Mathematician / Researcher

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in heitorbaldo

Academic Appointments

- 2024 – 2026 **Postdoctoral Research Fellow**, *University of São Paulo*, Brazil.
- 2024 – 2025 **Visiting Postdoctoral Research Fellow**, *Leipzig University*, Germany.
- 2019 – **Affiliate Researcher**, *IGDORE Institute*, Sweden.

Education

- 2019 – 2024 **Ph.D. in Bioinformatics (Mathematical Neuroscience)** , *University of São Paulo*, Brazil.
Thesis: *Towards a Quantitative Theory of Digraph-Based Complexes and its Applications in Brain Network Analysis*, advised by Koichi Sameshima and co-advised by André Fujita.
- 2014 – 2016 **M.S. in Applied and Computational Mathematics**, *University of Campinas*, Brazil.
Thesis: *Álgebras de Clifford e de Cayley-Dickson*, advised by Jayme Vaz Jr.
- 2009 – 2013 **B.S. in Mathematics**, *University of Campinas*, Brazil.

Publications

Preprint Manuscripts

- 2026 **Baldo, H., Baccalá, L., Fujita, A., & Sameshima, K.**, *Directed Q-Analysis and Directed Higher-Order Connectivity on Digraphs: A Quantitative Approach*, arXiv:2605.14178, <https://doi.org/10.48550/arXiv.2605.14178>
(preprint - submitted to a peer-reviewed journal)

Under Review Manuscripts

- 2026 **Baldo, H. & Fujita, A.**, *A Temporal Hypergraph Motif Framework for the Analysis of Higher-Order Dynamics of Epileptic Brain Networks*, (under review).

Unpublished Manuscripts and Academic Writings

- 2024 **Baldo, H.**, *Notes on Multi-Agent Reinforcement Learning*, (study notes).
- 2023 **Baldo, H.**, *Notes on Simplicial Neural Networks for Path Complexes*, (research notes).
- 2022 **Baldo, H.**, *Notes on Collapses and Persistent Path Homology*, (unpublished manuscript).

Open-Source Software

Since 2022 **Research Packages: DigplexQ, TemporalHypermotifs, PyTropical, Pytroids**, <https://github.com/heitorbaldo?tab=repositories>

- Open-source Python packages providing computational tools for research in directed higher-order network analysis and directed Q-analysis, temporal hypergraphs, tropical mathematics, and graphic matroid theory. Tools: [Python](#), [Sphinx](#), [PyPi](#), [NumPy](#), [SciPy](#), [Pandas](#), [NetworkX](#), [HyperNetX](#), [XGI](#).

Talks

- 2025 “Spectral Density of Multilayer Hypergraphs and Applications” in *6th Central German Meeting on Bioinformatics*, Leipzig University.
- 2022 “Graph Cellular Automata and Beyond: Applications in Network Neuroscience.” Institute of Mathematics and Statistics - USP. ([slides](#))
- 2022 “Path Homology Theory of Digraphs and Its Applications to Brain Network Analysis” in *Universität Leipzig - USP Workshop 2022* at the Institute of Mathematics and Statistics - USP. ([slides](#))

Teaching Experiences

- Spring 2021 **Teaching Assistant**, University of São Paulo, Brazil.
Course: *Multivariate Data Analysis* (MAE0330).
- Winter 2017 **Lecturer**, University of Campinas, Brazil.
Course: *Complex-Valued Neural Networks* (Minicourse).
- Spring 2016 **Teaching Assistant**, University of Campinas, Brazil.
Course: *Linear Algebra* (MA327).
- Winter 2016 **Lecturer**, University of Campinas, Brazil.
Course: *Bio-Inspired Algorithms: An Introduction* (Minicourse).

Technical Skills

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|----------------|---|
| Programming | Python • Julia • R • C/C++ • Rust • SQL • TypeScript • HTML/CSS |
| ML/DL | PyTorch • TensorFlow • scikit-learn • XGBoost • PyTorch Geometric • DGL |
| Backend & APIs | FastAPI • Pydantic • Uvicorn • PostgreSQL • Redis • SQLAlchemy • REST |
| Frontend | Streamlit • Gradio • React • Next.js • TypeScript |
| MLOps & Infra | Docker • MLflow • Weights & Biases • DVC • Evidently AI • Prometheus • Grafana |
| Math & Neuro | SageMath • Mathematica • MATLAB/Octave • NetworkX/HyperNetX/XGI • Gudhi • Giotto-TDA • EEGLAB • Brainstorm • FreeSurfer • SPM • FieldTrip • fMRIPrep • Brian2 • MNE |
| Others | Git • Linux • pytest • Sphinx • $\text{\LaTeX}$ |

Selected Participation in Events

- 2026 Homotopy Theory, K-theory, and Topological Data Analysis. *Fields Institute*.
- 2025 Thirteenth Symposium on Compositional Structures (SYCO 13). *University of Cambridge*.

- 2025 Workshop “Biology and Geometry”. *Max Planck Institute for Mathematics in the Sciences.*
- 2025 6th Central German Meeting on Bioinformatics. *Leipzig University.*
- 2025 40th TBI Winterseminar in Bled. *Bled, Slovenia.*
- 2024 Twelfth Symposium on Compositional Structures (SYCO 12). *University of Cambridge.*
- 2021 Workshop on Algebraic Graph Theory and Quantum Information. *Fields Institute.*
- 2021 4th Workshop on Algebraic Graph Theory and its Applications. *Mathematical Center in Akademgorodok.*
- 2020 XLIII Annual Meeting of the Brazilian Society for Neuroscience and Behavior. *Online.*
- 2020 Seminars on Probability and Stochastic Processes. *University of São Paulo.*
- 2017 6th Brazilian Conference on Intelligent Systems. *Federal University of Uberlandia.*
- 2016 II Brazilian Congress of Young Researchers in Pure and Applied Mathematics. *University of Campinas.*
- 2015 IV School and Workshop on Lie Theory. *University of Campinas.*
- 2014 Workshop Many Faces of Distances. *University of Campinas.*

Certifications & Training

- Spring 2026 Introduction to Agent-Based Modeling. *Santa Fe Institute.* [[Certification](#)]
- Spring 2018 Statistical Learning. *Stanford Online – Stanford University.* [[Certification](#)]
- Spring 2017 Introduction to Complexity. *Santa Fe Institute.* [[Certification](#)]
- Spring 2017 Minicourse on Machine Learning for Many-Body Physics. *IFT-Unesp.*

Languages

- **Portuguese:** Mother Tongue.
 - **English:** Advanced Level (C1*).
 - **German:** Pre-intermediate Level (A2*)
- (*) *Common European Framework of Reference (CEFR) level.*

Last updated August 02, 2026.